# Deep Learning Based Medical Image Analysis For Early Disease Detection

## Testing Report

Phase III project submission | 30 October 2026

Student / institution: enter the approved cover details. No identity or approval has been invented.

## Objectives and environment

Tests cover metadata and data loading, preprocessing, split integrity, model initialization, metrics, checkpoint save/load, end-to-end training mechanics, real dataset audit and model-backed inference. Windows, Python 3.11, PyTorch 2.15.0+cu132, torchvision 0.29.2+cu132, RTX 3050 Laptop GPU (CUDA).

## Executed test cases

| ID | Test scope | Expected result | Actual result | Status |
|---|---|---|---|---|
| T-01 | Fixture metadata adapter | Validated records and labels | Passed | PASS |
| T-02 | Grouped and predefined split | No leakage; official test preserved | Passed | PASS |
| T-03 | Image transform | Consistent tensor shape | Passed | PASS |
| T-04 | Model forward/checkpoint reload | Correct output shape | Passed | PASS |
| T-05 | Metric calculation | Finite metrics and confusion matrix | Passed | PASS |
| T-06 | One-epoch fixture pipeline | Train and checkpoint execute | Passed | PASS |
| T-07 | ISIC source audit | Readable data, manifest and duplicates checked | 1,275 records; no unreadables/missing/duplicates | PASS |
| T-08 | GPU model prediction | Checkpoint loads and gives probabilities | ISIC held-out image predicted; probabilities returned | PASS |
| T-09 | Grad-CAM inference | Valid overlay dimensions | 224x224 map; source-sized overlay | PASS |
| T-10 | Streamlit inference and state | Result persists and Grad-CAM renders | Verified in browser | PASS |
| T-11 | Full pytest suite | All tests pass | 13 passed in 6.99 s | PASS |
| T-12 | Compilation / dependencies | No compile or dependency errors | compileall passed; pip check clean | PASS |
| T-13 | NIH and APTOS full-data tests | Audit and train | Not executed: source datasets unavailable | PENDING |

## Method and summary

Synthetic fixtures validate software mechanics only; they are not medical evidence. The real ISIC audit, training, held-out evaluation, CUDA inference, app prediction and Grad-CAM were separately exercised. NIH/APTOS integration remains untested until the full sources are available. Final command: `.venv/Scripts/python.exe -m pytest -q

--basetemp=.pytest-tmp` (13 passed in 6.99 s). Bytecode compilation succeeded and `pip check` reported no broken requirements.

Known outstanding deliverables: public deployment, NIH/APTOS source data, and a recorded .mov demo. A concise recording script and checklist are provided.